

AI Fairness in Educational Systems: Evaluating Bias and Mitigation in Student Outcome Prediction

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Abstract— Machine learning systems are increasingly used in educational settings to predict student outcomes such as academic success and performance levels. However, these systems may reflect or amplify existing inequalities present in the data.

This study evaluates algorithmic fairness in predicting student outcomes using the Student Performance dataset. Two protected attributes: gender and age are analysed across two outcome definitions: pass/fail and high performance. Fairness is measured using Disparate Impact (DI) and Statistical Parity Difference (SPD), and a pre-processing bias mitigation technique is applied to assess its effectiveness.

The analysis shows that standard bias mitigation techniques may not significantly improve fairness and, in some cases, can worsen disparities across certain groups. The findings highlight the limitations of commonly used fairness metrics and demonstrate how outcome definitions influence fairness evaluation. The study emphasizes the need for deeper, system-level approaches to fairness beyond statistical adjustments.

1 INTRODUCTION

As machine learning systems are increasingly integrated into decision-making processes in domains such as education, concerns around fairness and bias have become more prominent. Predictive systems that assess student performance can influence access to opportunities, interventions, and institutional decisions. However, these systems rely on historical data, which may already contain structural inequalities.

Fairness in machine learning is often evaluated using statistical metrics that compare outcomes across different groups. While these metrics provide a way to detect disparities, they do not necessarily resolve the underlying causes of bias. Additionally, fairness outcomes can vary depending on how success is defined, how data is distributed, and how mitigation techniques are applied.

This study examines fairness in an educational prediction context by analyzing how protected attributes influence outcomes and how bias mitigation techniques affect fairness metrics. The goal is not only to measure fairness but to understand the limitations of current approaches when applied to real-world datasets.

2 METHODOLOGY

This study follows a structured approach to evaluate fairness in a supervised learning setting using an educational dataset.

2.1 Dataset and Variables

The analysis is conducted on the Student Performance dataset, which includes demographic, social, and academic variables for 649 students. Two outcome variables are defined:

- Pass/Fail (threshold-based classification)
- High Performance (higher threshold classification)

Two protected attributes are selected:

- Gender (Male, Female)
- Age (Younger, Older)

2.2 Fairness Metrics

Two widely used fairness metrics are applied:

- Disparate Impact (DI): Measures the ratio of favourable outcomes between unprivileged and privileged groups

- Statistical Parity Difference (SPD): Measures the difference in favourable outcome rates between groups

These metrics are used to evaluate whether outcomes are distributed equitably across groups.

2.3 Bias Mitigation Approach

A pre-processing bias mitigation technique (rebalancing through resampling) is applied to reduce disparities in group representation. The dataset is transformed by balancing the representation of gender groups.

2.4 Model Evaluation

A classification model is trained on both the original and transformed datasets. Fairness metrics are computed:

- before mitigation
- after dataset transformation
- after model training

The goal is to evaluate whether mitigation techniques improve fairness outcomes at different stages of the pipeline.

3 DATASET AND ANALYSIS

3.1 Dataset Selected

The dataset selected for this project is the Student Performance Dataset (Portuguese language).

Link: <https://www.kaggle.com/datasets/uciml/student-alcohol-consumption/data>

3.2 Domain

The dataset belongs to the Education domain, as it focuses on student academic performance and outcomes.

3.3 Observations

The dataset contains 649 observations (students).

3.4 Variables

The dataset contains **33 variables**, including demographic, social, and academic performance-related features.

3.5 Variable Selection

Two dependent variables were selected:

1. **Pass/Fail** (derived variable)
 - Defined as: Pass = 1 if $G_3 \geq 10$, else Fail = 0
2. **High Performance** (derived variable)
 - Defined as: High = 1 if $G_3 \geq 15$, else Low = 0

These variables represent favourable and unfavourable academic outcomes.

3.6 Protected Class

Two protected class variables were selected:

1. Sex (Gender)
 - Values: Male (M), Female (F)
2. Age
 - Converted into groups:
 - Younger (<18)
 - Older (≥ 18)

3.7 Legal Precedence for Protected Class

Gender (Sex): Protected under laws such as Title IX and general equal protection principles that prohibit discrimination based on gender in education.

Age: Protected under Age Discrimination laws, which aim to prevent unfair treatment based on age.

4 DATASET EXPLORATION

4.1 Protected Class and Sub Groups

The protected class variables and their subgroups are:

Table 1 — Protected Class and their Sub Groups

Protected Class	Sub Groups
Gender	Male (M), Female (F)
Age	Younger (<18), Older (>=18)

4.2 Discretization

The subgroups were converted into numerical values as follows:

Table 2 — Discretization of sub groups

Protected Class	Sub Groups	Numerical Values
Gender	Male (M), Female (F)	M=0, F=1
Age	Younger (<18), Older (>=18)	Younger=0, Older=1

4.3 Protected Class

The two protected classes selected for further analysis are **Gender (sex)** and **Age group**.

4.4 Frequency Tables

Frequency tables for each protected class documenting number of members in each sub group associated with each dependent variable.

Table 3 — Gender vs Pass

Pass	0	1
Gender		
0	50	216
1	50	333

Table 4 — Age vs Pass

Pass	0	1
Age		
0	62	406
1	38	143

Table 5 — Gender vs High Performance

High_perf	0	1
Gender		
0	223	43
1	295	88

Table 6 — Age vs High Performance

High_perf	0	1
Age		
0	376	92
1	142	39

4.5 Bar Charts for Protected Class

Bar charts for protected class sub groups as a function of the dependent variable.

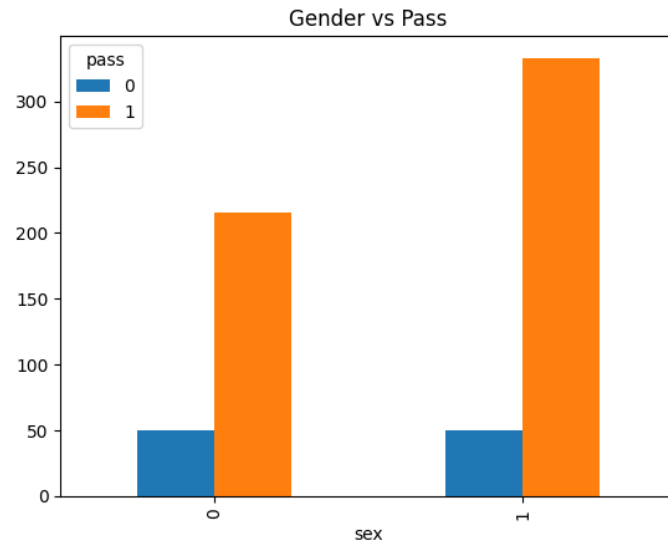


Figure 1— Gender vs Pass

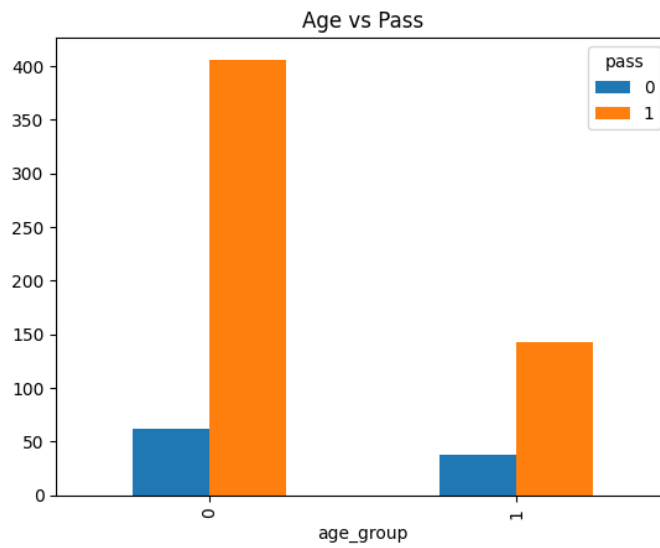


Figure 2— Age vs Pass

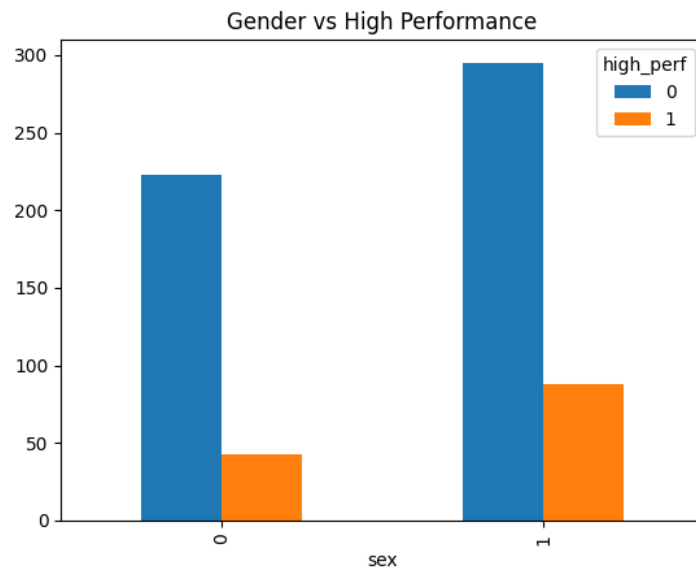


Figure 3— Gender vs High Performance

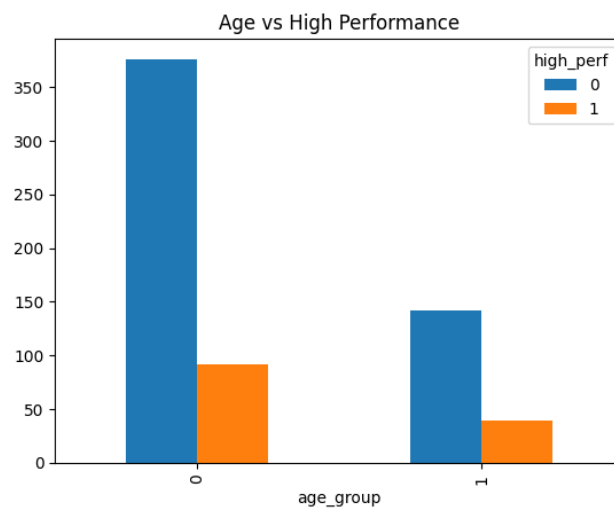


Figure 4— Age vs High Performance

5 FAIRNESS METRIC SELECTION AND MITIGATING BIAS WITH PREPROCESSING ALGORITHMS

5.1 Privileged and Unprivileged Groups

The protected classes selected in this dataset are gender (sex) and age group.

For gender,

- Privileged group: Male (0)
- Unprivileged group: Female (1)

For age group,

- Privileged group: Older students (≥ 18 , encoded as 1)
- Unprivileged group: Younger students (< 18 , encoded as 0)

These groupings are defined based on the assumption that historically, certain groups may have advantages in educational outcomes.

5.2 Fairness Metrics Table

The two fairness metric algorithms selected for this analysis are Disparate Impact (DI) and Statistical Parity Difference (SPD).

Disparate Impact (DI) measures the ratio of the probability of a favourable outcome for the unprivileged group to that of the privileged group. It is defined as:

$$DI = P(\text{favourable outcome} \mid \text{unprivileged group}) / P(\text{favourable outcome} \mid \text{privileged group})$$

An ideal value of DI is 1, which indicates that both groups receive favourable outcomes at equal rates. A value greater than 1 indicates an advantage for the unprivileged group, while a value less than 1 indicates an advantage for the privileged group.

Statistical Parity Difference (SPD) measures the difference in the probability of a favourable outcome between the unprivileged and privileged groups. It is defined as:

$SPD = P(\text{favourable outcome} \mid \text{unprivileged group}) - P(\text{favourable outcome} \mid \text{privileged group})$

An ideal value of SPD is 0, which indicates equal treatment across groups. A positive value indicates an advantage for the unprivileged group, while a negative value indicates an advantage for the privileged group.

These metrics were selected because they directly compare outcome distributions across groups and are widely used for detecting bias in classification tasks.

Table 7 — Fairness Metrics Table

Protected Class	Outcome Variable	Disparate Impact (DI)	Statistical Parity Difference (SPD)
Gender	Pass	1.0707	0.0574
Gender	High Performance	1.4213	0.0681
Age	Pass	1.0981	0.0775
Age	High Performance	0.9123	-0.0189

5.3 Bias Mitigation Technique

The bias mitigation technique selected for this project is rebalancing through resampling, which is a pre-processing method. This approach involves creating a balanced dataset by ensuring that the privileged and unprivileged groups have equal representation.

In this case, the dataset was balanced based on the protected class gender, by randomly sampling an equal number of instances from both male (privileged) and female (unprivileged) groups. This transformation was performed with respect to the dependent variable Pass/Fail, as it represents the primary outcome of interest.

The purpose of this method is to reduce bias by eliminating any imbalance in group representation, which may otherwise influence the model or fairness metrics.

5.4 Fairness Metrics Table – Transformed Dataset

Table 8 — Fairness Metrics Table – Transformed Dataset

Protected Class	Outcome Variable	Disparate Impact (DI)	Statistical Parity Difference (SPD)
Gender	Pass	1.0694	0.0564
Gender	High Performance	1.3256	0.0526
Age	Pass	1.1049	0.0819
Age	High Performance	0.8826	-0.0241

The fairness metrics computed on the transformed dataset show only minor changes compared to the original dataset. For gender, the Disparate Impact for the High-Performance outcome decreased slightly from 1.4213 to 1.3256, indicating a small reduction in the advantage for the unprivileged group. Similarly, the Statistical Parity Difference also decreased slightly.

However, for the Pass outcome, the values remained almost unchanged, suggesting that the bias mitigation technique had minimal effect. For the age-based analysis, the fairness metrics showed slight variations, but no significant improvement in fairness was observed.

Overall, the results indicate that while the rebalancing technique adjusted the dataset distribution, it did not substantially improve fairness metrics. This suggests that the original dataset was already relatively balanced, and simple resampling was not sufficient to significantly alter fairness outcomes.

6 MODEL TRAINING AND FAIRNESS EVALUATION

6.1 Change in Fairness Metrics

Table 9 — Fairness Comparison (Test Data)

Protected Class	Metric	Original Test	Transformed Test
Gender	DI	1.0484	1.0950
Gender	SPD	0.0412	0.0753
Age	DI	0.9687	1.0701
Age	SPD	-0.0281	0.0555

Table 10 — Change Analysis

Metric	Change After Transformation	Change after Model
Gender DI	Negative Change	No Change
Gender SPD	Negative Change	No Change
Age DI	Positive Change	No Change
Age SPD	Positive Change	No Change

The fairness metrics computed on the testing dataset show that the impact of bias mitigation varies across protected classes. For gender, both Disparate Impact and Statistical Parity Difference moved further away from their ideal values after transformation, indicating a negative change in fairness. This suggests that the rebalancing technique slightly increased disparity between male and female groups.

In contrast, for the age-based protected class, both fairness metrics moved closer to their ideal values. The Disparate Impact shifted closer to 1 and the Statistical Parity Difference moved closer to 0, indicating a positive change in fairness. This

shows that the bias mitigation technique was more effective for age than for gender.

After training the classifier, the fairness metrics remained largely consistent with the dataset-level results, indicating no significant additional impact from the model. This suggests that bias present in the dataset persists through the model, and that simple pre-processing techniques may not uniformly improve fairness across all groups.

7 CONCLUSION

This study highlights that evaluating fairness in machine learning systems is not a straightforward task and cannot be resolved through simple statistical adjustments. While fairness metrics such as Disparate Impact and Statistical Parity Difference provide useful indicators of disparity, they do not capture the full complexity of bias present in real-world systems.

The results show that pre-processing techniques such as rebalancing may not significantly improve fairness and can produce uneven effects across different protected groups. In this study, fairness improved slightly for age-based groups but worsened for gender in certain cases, demonstrating that mitigation strategies are not universally effective.

Additionally, the findings indicate that bias present in the dataset persists through the model, as training did not significantly alter fairness outcomes. This suggests that addressing fairness requires deeper intervention beyond data balancing, including reconsideration of feature selection, model design, and outcome definitions.

Finally, the analysis shows that fairness is sensitive to how outcomes are defined. Different definitions of success, such as pass versus high performance, lead to different fairness results, highlighting that fairness is not only a property of the model, but also of how the problem itself is framed.

Overall, this study underscores the need for more robust and context-aware approaches to fairness that go beyond standard metrics and consider the broader system in which these models operate.